AI for Good Sandbox Hackathon – Team Sankalp – HISN: the law layer and last mile for agentic government

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Github link: https://github.com/Eswarpnbs/HISN (Apache-2.0; run python selftest.py and python tools/verify_all.py to re-derive every figure)

1. Introduction

The UAE plans to deliver half of its government services through AI agents within two years (Cabinet, 23 April 2026). A national lab tests whether AI models and agents are secure (UAE Cyber Security Council, May 2026), and a planned control tower will log what government agents do and keep them within their authority (Minister of State for AI, 14 October 2026). Two questions remain: is an action lawful, under which clause, and did the citizen actually receive the outcome? HISN complements these systems as the law layer and the last mile: no essential message lost in silence, no rule issued untested. In this use case an operator's AI SMS filter is tightened under pressure after a scam wave, and a genuine, authenticated hospital message is silently dropped while look-alike spoofs probe the same filter. Nobody can say who may approve a filter change or who answers for the loss. HISN, a working open-source prototype, governs those decisions: a deterministic policy node blocks only on verbatim, hash-pinned UAE legal clauses; institutional agents (TDRA, Health Authority, Federal AI and Data Authority) propose actions on retrieved evidence; a named human decides; and every step is sealed in a hash-chained audit log.

HISN covers the four agent roles in the brief: a domain solution (the governed SMS firewall), a policy and legal adviser, institutional participants, and multi-agent coordination (the Y.3172 MLFO releases events progressively, so each agent sees only its own events, at its own time). The same engine runs on UAE, India and Singapore by swapping the knowledge base and rule pack, without code changes. We map it to ITU-T Y.3172 (all nine nodes) and the AI Readiness Framework 2.0 (metric IDs checked against ITU's dimensions.json). Each node is grounded in UAE instruments: a knowledge base of 84 hash-pinned official records (35 UAE instruments in English and 29 in authoritative Arabic, 10 ITU-T Recommendations including the X.1231-X.1247 anti-spam series, and 10 comparator instruments) and 38 verbatim clauses, with India and Singapore packs to test portability. From the mapping we derive seven domain-level policy gaps (G1 to G7) with verified global examples. Scenario messages are synthetic; four tests use real public data (5,574 and 5,971 real SMS, 13,240 real Arabic messages, the UAE Government's own descriptions of its laws). On every real SMS text the filter holds (275), the system released none of 157 real scams.

2. Description of the use case

The problem is that essential messages (a critical lab result, an appointment change, a traffic-fine receipt) can be lost to an AI filter without anyone knowing, and the rules around that filter are unclear. Existing solutions score messages on keywords, links and sender look-alikes and drop what looks like spam, with no record of why. The pressure that tightens filters is real: the UAE Cyber Security Council reported fraudulent messages up about 35% in a year (WAM, 1 Feb 2026), and in April 2026 the Public Prosecution acted against a cross-emirate network sending fake toll-fine SMS (35 reports, 29 in Dubai). We found no instrument that gives government or health senders protected status (G2), covers fraud and impersonation in filtering duties (G1), assigns authority to approve or override changes to an operator's AI filter (G5), or allocates scam losses when those duties fail (G6). Language-model assistants that explain the law add a further risk: in our tests an LLM alone made 41 unsupported legal citations on 33 items.

HISN does three things beyond checking AI against existing rules. (1) The right to be reached: essential senders are never silently dropped; a held message goes to a named officer within a clinical time limit, the sender gets an accountable receipt (real SMPP 3.4 server), and every run measures both harms, essential messages lost and scams delivered, as a MIZAN balance. (2) Crash-test the rule: a regulatory wind tunnel tests any draft rule or filter change before issuance against verified law, the hierarchy of

norms and nine public-interest invariants, and a gap engine re-runs the scenario with candidate provisions to show which rules carry the load. (3) Declared autonomy, enforced: each component's ITU-T Y.3173 level is declared, observed and enforced, and a change that breaks it is held; named approval plus a second reviewer within a time limit work as maker and checker. Underneath: a policy node that blocks only on a verified binding clause, agents that only propose (six fixed gates), named sign-off and a hash-chained audit log.

3. Use case requirement

Each node of the pipeline has requirements, marked critical (C), expected (E) or optional (O).

SRC: C: sender identity comes from the authenticated SMPP bind, never from message text; scenario messages are synthetic with fictional senders; public real SMS texts are used only for evaluation. E: hospital messages carry a FHIR priority (routine, urgent, asap, stat). O: network stressor data (rain-driven link failures).

C and PP: C: every event and decision sealed in the hash-chained audit log; each agent sees only events addressed to its role and already released; the clinical tier comes only from an authenticated sender's declared priority, never from content; each agent decision is logged with its assessment, retrieved evidence, cited clauses and stated uncertainty (a reviewable rationale, not raw model reasoning). E: a knowledge base of official records, SHA-256 pinned and checkable by anyone.

M: C: the language model only proposes; a decisive action needs a cited source that is binding and within the agent's competence, otherwise a gap is logged. E: pre-registered held-out tests and four baselines (fixed rules, LLM only, LLM with knowledge base, full HISN).

P: C: block only on a verified binding clause; offshore health-data transfers blocked (Federal Law 2/2019 Art. 13); no legitimate essential message silently lost; uncovered events logged as gaps. E: the same rules run on UAE, India and Singapore packs.

D and SINK: C: a held essential message reaches a named officer, with a second reviewer if it is unresolved at the tier limit (5, 15 or 60 minutes; provisional, pending clinician review); the sender gets a receipt with reason and deadline. E: a daily assurance page for the regulator; a citizen verifier in six languages (three reviewed).

MLFO and Sandbox: C: every draft rule and filter change tested before issuance; emergency exceptions must expire; sign-off refused without evidence. E: gap-engine measures of protection, access and MIZAN. O: extrapolation from the UAE to India and Singapore.

4. Mapping with ITU-T Y.3172 Readiness 2.0 Report, and domain-specific policies

	Use case (HISN implementation, data and processing)	Readiness 2.0 Report	Documents (policies, strategies, regulations)
SRC	SMS over SMPP 3.4; hospital messages as FHIR CommunicationRequest; sender identity from the bound account. Data: synthetic scenario SMS (91 rows, fictional senders, CC BY 4.0); public real SMS for evaluation only; network stressor proxy from the Turkcell 2021 ITU AI/ML in 5G Challenge (985,308 link-days; aggregate profile only; not UAE data).	D1, D4, D13: Area Coverage (d13_m08), Network Quality (d13_m09); open data and open-source management	Federal Law 2/2019 Art. 1 (circulation of health information); TDRA UEC Policy cl. 8.8 (marketing sender label only); India TRAI Government Message definition (IN-TRAI-GOVMSG)
C	Time-gated, role-scoped event bus; hash-chained audit log; SMPP bind and authentication. Agents see only events addressed to their role.	D1, D3, D13: Integrated Workflow (d3_m01); human interaction and supervision	Federal Law 2/2019 Art. 13 (health data stays in the State)
PP	Sender authentication and route checks; clinical tier from declared priority. Knowledge base: 84 official records (uaelegislation.gov.ae, itu.int, regulators; URLs in kb/manifest.csv), SHA-256 pinned and identified from their own first page; Arabic texts read by OCR (unreviewed; all 1,443 English article numbers found); 38 verbatim clauses via ITU's unmodified ingest pipeline (36 exact, 2 layout-reconstructed and declared). No personal data: synthetic scenarios, public CC BY 4.0 evaluation data, aggregate network data.	D1, D2, D3: Coverage and Scale (d3_m07: 6 jurisdictions plus ITU-T, 84 records, 38 clauses); contextualisation and regional development	Federal Law 2/2019 Art. 16 (consent exceptions)
M	SMS filter scoring; institutional LLM agents with	D2, D3, D5, D9: generative	TDRA UEC Policy cl. 1.10,

	retrieval (qwen2.5:14b); policy and legal adviser; four baselines. Agents only propose; six deterministic gates (evidence, mandate, binding basis, competence, self-consistency, applicability: a release needs an authenticated sender); fixed message checks in English and Arabic given to the model as facts; majority vote of 3; invented citations rejected.	AI content ecosystem (LLMs only propose); model and dataset metadata (agent cards with ITU-T Y.3176 fields, dataset datasheet); no metric ID claimed	6.1, 11.1; Cybercrime Law Art. 40 (internet fraud, false impersonation)
P	Deterministic policy gateway; UAE, India and Singapore rule packs; clinical tiers; wind-tunnel invariants I1 to I9. Blocks only on verified binding clauses; soft law never blocks; uncovered cases logged as gaps; drafts tested before issuance.	D7, D8, D10: Safety Policies (d10_m06), Explainability (d10_m07), Audit (d10_m09), Data Sovereignty (d10_m10), Intent Representations (d7_m03), International Cooperation (d7_m08)	Telecom Law Art. 12, 14(3); Federal Law 2/2019 Art. 13, 16; TDRA UEC cl. 11.1 to 11.2, 13; PDPL Art. 2(2); UAE AI Charter (soft law)
D	Deliver, hold or block decisions; SMPP responses (block = 0x45) and accountable receipts with reason, deadline and audit hash; escalation to the named authority. Essential messages are never silently dropped.	D5, D6, D12: human oversight and inclusion; no metric ID claimed	TDRA UEC Policy cl. 11.1 to 11.2; Telecom Law Art. 14(3) (subscriber service standards)
SINK	Citizen and hospital; named officers and duty supervisor; regulator outputs (daily assurance page, impact cards, gap register, ITU kb_decisions); citizen verifier in six languages (three reviewed, advisory only).	D5, D6, D12, D13: Uptime (d13_m07, 0 silently lost in the daily assurance run); collaboration between AI and human	UAE AI Charter (soft law, human oversight)
MLFO	Coordinator: simulated clock, dispatch, referral routing between authorities (routed once, audited), tier-based second review, emergency state and expiry; one-command reproducibility (selftest.py); CI wind tunnel on pull requests.	D5, D7, D8: Coordination Level (d7_m01, strategy alignment across institutions), Interoperability Standards (d7_m04: SMPP 3.4, HL7 FHIR, Y.3172 export)	Federal Law 2/2019 Art. 13, 24
Sandbox	Governed vs ungoverned twin; gap engine; regulatory wind tunnel; fairness check; filter-change check; synthetic scenarios, real public data for evaluation.	D10, D11: Policy Sandbox (d10_m01), Policy Evaluation (d10_m02), Policy Extrapolation (d10_m03), Language Repository (d11_m07)	No binding instrument (synthetic sandbox); ITU-T Y.3173 levels (docs/automation_levels.md)

Metric IDs are checked against ITU's simulation/data/framework/dimensions.json (30 Oct 2026). Regional Language Support (d11_m08) is not claimed: 29 Arabic legal texts and Arabic message checks are in, but native review is pending.

5. Error scenarios

Step 1: The governed SMS firewall and agents are deployed in the operator's network and trusted to protect citizens from SMS scams. Sender identity comes from the SMPP account, not from the message.

Step 2 (scam wave): the operator tightens the filter threshold from 0.5 to 0.2 under pressure. A genuine, authenticated hospital message (an appointment change) now scores above it. Ungoverned, it is silently dropped. With HISN it is quarantined and escalated to the Health Authority agent, which proposes release by a named officer with cited evidence; only a named human can release it.

Step 3 (offshore failover): an automatic failover tries to move health data abroad. The policy node blocks it on Federal Law 2/2019 Art. 13 (ungoverned: allowed). The wind tunnel tests the draft "during a declared emergency, message logs may be stored abroad for 72 hours": as a regulation it is DO NOT ISSUE AS DRAFTED (conflicts with Art. 13); as a Health Authority resolution, Art. 13's own route, it is READY FOR NAMED HUMAN REVIEW. An expiry probe checks that the exception ends (gap G4).

Step 4 (silent retuning): six months later a vendor retunes the threshold to 0.15 without telling the operator or regulator, and a genuine government message (a traffic-fine receipt) is held. The agents ask who may approve or override filter changes. No competence norm exists, so the code turns their answer into a logged gap (G5, discovered by the agents). The audit log records an ITU-T Y.3173 violation (demand mapping declared human, no approval); with enforcement on (run_demo.py --enforce-levels) both unapproved changes are held and both genuine messages delivered. Only an Ed25519-signed approval bound to that change and value, from a competent role, in its validity window and not from the requester unlocks a change; a typed name, an edited, copied or expired record, or a self-approval is held (run_approval_demo.py; demo keys). Headline figures use enforcement off.

Step 5 (where HISN can be challenged): (a) Spoofing: sender authentication plus a verified-sender lane fixes invariants I3, I7 and I9 with no regression and raises MIZAN from 76.7 to 94.2; a blunt fraud duty alone lowers it by 0.5. (b) The model can be wrong, so each failure became a fixed rule tested on a new set written before any run: held-out v4 found a spoof recommended for release, so gate 6 now requires an authenticated sender for any release; v6 found genuine messages that say "pay" held as scams, so identity and content are now separate (fixed message checks given to the model as facts). (c) Language fairness: a filter-change check blocks threshold changes that score languages unequally. (d) Cross-border: UAE and India packs agree on 56% of decisions, and a genuine foreign essential sender arrives like a spoof (G7).

Evidence (python selftest.py and tools/verify_all.py re-derive every row; "fixed" = gold answers or counting rules committed in git before the first scored run):

Test	Data	Result	Fixed
Agents on real texts	All 275 real SMS the filter holds (Mishra and Soni 2022; no sampling)	251/275 (87-94%); real scams released 0 of 157 (gate 6 stopped 4); 20 of 118 legitimate held for an officer	Yes
Agents, held-out v7 and v8	18 new messages, 6 in Arabic; v8 with majority vote of 3	v7 8/8, v8 10/10, 0 harmful (18/18, 82-100%)	Yes
Agents, earlier held-out first runs	v2-v6, paraphrase (n = 2-9 each)	v2 4/4, v3 1/2, v4 6/9 (1 harmful), paraphrase 6/8, v5 4/8, v6 5/8 (1 harmful); each led to a fix	Yes
Baselines on 33 gold items	Synthetic	HISN 26/33, 1 harmful, 0 unsupported citations; LLM only 16/33, 10 harmful, 41 unsupported; rules 23/33, 10 harmful; RAG 23/33, 9 harmful	Yes
Twin run	Governed vs ungoverned	Essential messages silently lost 10 to 0; offshore transfers allowed 2 to 0	n/a
Message checks, real SMS	5,971 real SMS (Mishra and Soni 2022)	Risk signal on 79% of real smishing and 0.1% of real legitimate SMS	Yes
Arabic checks, real messages	13,240 real Arabic messages (Zayed University)	0.8% of legitimate text flagged; none flagged as asking for a secret	Yes
Filter drift, real SMS	5,574 real SMS (UCI)	Threshold 0.5 to 0.2: legitimate blocked 0 to 103, spam caught 88	n/a
Retrieval, official wording	13 descriptions of UAE laws published on u.ae	Right law in top 5: 12/13; top 1: 3/13	Yes
Arabic OCR structure	29 Arabic legal texts	All 1,443 article numbers of the English texts found	n/a
SMPP interoperability	Independent client library (smpplib)	5/5 cases as expected; found and fixed one protocol error	n/a

Baseline accuracy intervals overlap, so we claim a difference in harm and citations, not accuracy. Repeated runs: 44/44 identical.

6. Policy gaps and standards gaps derived from the mapping exercise

Seven domain-level gaps came out of the mapping and the scenarios. G1 to G4 are verified within the instruments searched; G5 and G7 were discovered by the system; G6 is led by a comparator. These are domain-level findings, not claims that a country breaks its law, and we make no national recommendation.

Gap	Finding and evidence	Verified global example and what transfers	Policy or standards implication
G1	Telecom filtering duties cover marketing spam, not fraud or impersonation. The word "fraud" has zero hits in the TDRA UEC Policy (searched 26 Oct 2026); cl. 1.10, 6.1, 11.1; Cybercrime Law Art. 40.	Australia: Scams Prevention Framework Act 2025 and Industry Code C661:2022 put scam-prevention duties on telecoms. The design transfers; the legal setting differs.	Extend filtering duties to fraud and impersonation (legislative scope).
G2	Government and health senders lack protected status. Marketing senders carry the AD- label (cl. 8.8), but no instrument we found protects government or health messages from AI filtering; cl. 13; PDPL Art. 2(2).	India: TRAI regulations define Government Messages, sent through DLT with a "-G" header and not blockable through preferences. Australia: the SMS Sender ID Register keeps registered names (including government) and labels unregistered senders.	A protected sender register with authentication; in our gap engine it raised MIZAN from 76.7 to 94.2.
G3	AI scanning of health SMS for fraud has no explicit legal basis. Federal Law 2/2019 Art. 16 lists five exceptions; telecom fraud filtering is not one; PDPL Art. 2(2).	EU Regulation 2021/1232: a purpose-limited, temporary legal basis for voluntary scanning. Its subject (child sexual abuse material) differs, so only the legal-basis design transfers.	A narrow, purpose-limited legal basis; proposed for ITU standards work.

G4	No emergency path for systematic, AI-triggered failover of health data. Art. 13 violations carry AED 500,000-700,000 (Art. 24); the Executive Regulation lets a provider open one patient's file without consent, nothing more.	EU: GDPR Art. 49(1)(f) and EDPB Guidelines 2/2018: vital interests cover specific, occasional cases, not systematic transfers. The same gap exists there.	A pre-approved, time-limited emergency route, tested in the wind tunnel.
G5	No instrument assigns authority to approve, override or require protection in operators' AI filters. Discovered by the agents; general powers over subscriber service standards (Telecom Law Art. 14(3)) are not applied to AI filter changes.	Australia: Code C661 has blocked scam SMS reported to the ACMA (partial). No pre-deployment approval authority found in Australia or the EU.	An authority and an approval record for filter changes; proposed for ITU standards work (Y.3173: demand mapping = human).
G6	No allocation of scam losses to telecom operators when anti-scam duties fail; bank-side compensation exists (CBUAE Consumer Protection Regulation, Circular 8/2020, Art. 6).	Singapore Shared Responsibility Framework: the telco bears SMS-scam losses when it breached its filtering duties and the bank complied.	Define telecom-side responsibility.
G7	Essential-sender recognition does not cross borders. India identifies government messages domestically (DLT, "-G" suffix); the UAE recognises only marketing labels; a genuine foreign sender arrives like a spoof (measured).	Singapore Sender ID Registry (partial): overseas senders only through a Singapore entity; protection applies to Singapore numbers only; no mutual recognition found.	An open question for international coordination; needs expert review.

Standards gaps. (1) Policy as code: no standard format exists for machine-readable rules that say who may block, on what basis and with which evidence. Every HISN rule is a YAML record bound to a hash-pinned clause (e.g. HEALTH_LOCALISATION_01: health data to a destination outside AE, BLOCK_PROHIBITED, FEDERAL_LAW, UAE-FL-2-2019 Art. 13 with verbatim quote). We offer the schema to ITU. (2) Authority record for agent actions: no standard defines the record that proves a human approved an agent's change (G5); HISN joins the declared Y.3173 level, the signed approval and the cited binding clause in one record. (3) No standard for cross-border recognition of essential senders (G7). (4) None of the 111 metrics in ITU's dimensions.json measures whether an agent's consequential action had verified authority; we propose an agent-authority metric. All three proposals, and our fix for Arabic legal PDFs that ITU's ingest pipeline reads as broken text, are posted as issues #1-#3 on ITU's reference repository (github.com/CrashingGuru/ITUAIReadiness/issues).

Limits we state. Sets v2-v8 are synthetic and small; v7 and v8 were written by the team on the day of the rule they test. In the real set the texts are real but sender authentication is simulated (no public data has it; over SMPP it comes from the bound account). The real SMS are English research data and the Arabic data is tweets. Arabic legal texts are OCR, unreviewed for wording; 3 of 6 verifier languages are natively reviewed. Approvals use demo keys; the SMPP server is not a production SMSC; network data is a proxy.

Next steps. A 30-day pilot in the ITU Sandbox with one UAE operator and one hospital sender: real authenticated traffic, no message content stored, measuring essential messages lost, scams delivered and time to a named release. Also keys from the authority's identity service, native review of the Arabic texts, expert review of G3, G4 and G7. Reproduce: bash setup.sh, then python tools/verify_all.py and python selftest.py.